Tzchen (Edison) Wu

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Education

Bachelor of Science in Electrical Engineering, Minor in Japanese Dec 2027
University of Nevada, Las Vegas
Honors: Dean's Honor List – Spring 2021, Spring 2026

Projects

Car Turn Signal Controller – SystemVerilog, Quartus II, FPGA Sep 2024

- Designed a 7-state Moore FSM turn-signal controller for a 1965 Ford Thunderbird in SystemVerilog
- Simulated and verified FSM operation and output waveforms using ModelSim
- Implemented the design in Quartus II on a **DE2-115 FPGA** and verified operation on physical hardware

USB Power Supply & Charging Circuit Apr 2026

- Designed and prototyped a regulated 5V USB power supply using an **LM7805,** validating it in LTspice before breadboard testing and perfboard soldering
- Verified voltage and current on both using a **Keithley 2110 digital multimeter**
- Implemented USB charging and verified operation by charging iPhones and other USB devices

Eddie Calendar – iOS Calendar Application May 2026 – Present

- Developing a native Swift iOS calendar integrating **3 event sources –** Google Calendar, Apple Calendar, and local events
- Integrated external data from Apple WeatherKit and a news API to generate daily look-ahead notifications
- Built 4 custom calendar views with filtering and customizable visibility

Personal AI Location Server – Ubuntu, Docker, Dawarich, Tailscale, Ollama Mar 2026

- Designed and deployed a self-hosted, containerized server to ingest, store, and visualize iPhone GPS data privately
- Integrated **Ollama** for local natural-language querying and analysis of location history
- Configured a **Tailscale VPN** for secure access and troubleshot networking and mobile-to-server integration

Work Experience

UGym Technology Corp. – CES Sales Representative Jan 2023 & Jan 2024

- Represented UGym Technology at 2 CES exhibitions with real-time Mandarin–English interpretation
- Demonstrated products, explained features, answered questions, and spoke with prospective customers

Surglasses – Engineering Intern Feb 2022 – Mar 2022

- Soldered wires and connectors, applied heat-shrink, and followed schematics to assemble wiring
- Revised the English-language user manual for an electronic product originally written in Mandarin

Point Robotics Medtech Inc. – Engineering Intern Jun 2021

- Modeled parts in **SolidWorks**, prepared them for **3D printing**
- Translated and revised a company press release from Mandarin to English

Leadership

Founder & President, UNLV Run Club Aug 2026 – Present

- Grew weekly group runs to 5 participants and arranged peer support for run leadership

UNLV Marching Band – Trombone Section Leader Aug 2024 – May 2026

- Coordinated 19 trombone members, including 9 freshmen, and split the section into 2 balanced groups
- Led sectionals, tracked attendance, conducted uniform checks, and communicated expectations to the section

Volunteer work

UNLV Marching Band

- Participated in community outreach performances and events supporting local schools and nonprofit organizations

Arbor View High School Track Meet

- Assisted with event operations and athlete support during a high school track meet

Technical Skills

UC OASiS Foundations – Semiconductor Processing & IC Fabrication
Programming: SystemVerilog, C, C++, Swift
Software & Tools: Quartus II, ModelSim, LTspice, MATLAB, Arduino, Git, Linux/Ubuntu, Docker, SolidWorks
Hardware & Test: FPGA, breadboarding, soldering, oscilloscopes, function generators, DMMs, 3D printing
Languages: English (Native), Mandarin (Native), Japanese (Conversational)